

How gamma controls readout learning

An attainable target can be slow to learn because the needed kernel eigenvalues are small relative to the largest one. Gamma controls these relative rates through an explicit smoothing operator. Our theorem bounds how much output error must remain after any number of frozen-feature GD updates. Its prediction substitutes gamma-dependent upper bounds on the learning rates into the GD error formula. The argument specializes the [V4 ratio theorem](#), on the same geometry as our executed GD and Adam experiments.

Notation. Eigenvalues use μ ; all norms are Euclidean.

$J_\gamma, K_\gamma = J_\gamma J_\gamma^T$	Normalized features and readout kernel
$\mu_i, u_i; \rho_i = \mu_i/\mu_1$	Descending eigenpairs; relative learning rates
$y, r_n; p_i = u_i^T y ^2/\ y\ ^2$	Target, residual; target energy in mode i

1 The rate that matters is relative to the largest eigenvalue

Freeze W equally spaced centers c_j , spacing h , and a common slope γ . On m input samples x_a , train readout w from zero:

$$J_\gamma[a, j] = \frac{\tanh(\gamma(x_a - c_j))}{\sqrt{m}}, \quad J_\gamma[a, 0] = \frac{1}{\sqrt{m}}, \quad y_a = \frac{f(x_a)}{\sqrt{m}}.$$

For loss $\frac{1}{2}\|J_\gamma w - y\|^2$, GD gives $r_{n+1} = (I - \eta K_\gamma)r_n$. The largest eigenvalue limits the stable shared step: $0 < \eta < 2/\mu_1$. At $\eta = 1/(2\mu_1)$, the exact relative error is

$$e(n)^2 := \frac{\|r_n\|^2}{\|y\|^2} = \sum_i p_i(\gamma)(1 - \rho_i(\gamma)/2)^{2n}.$$

A ratio of 10^{-6} needs about two million updates for one e-fold reduction of that component. The target weights determine which rates matter. A positive eigenvalue describes a representable component, however slow its decay; a zero eigenvalue contributes a fixed error floor.

2 Gamma enters the spectrum through smoothing

A tanh is a smoothed step: $\tanh(\gamma \cdot) = s_\gamma * \text{sign}$, where $s_\gamma(x) = \frac{\gamma}{2} \text{sech}^2(\gamma x)$. Its Fourier multiplier is

$$M_\gamma(\omega) = \frac{z}{\sinh z}, \quad z = \frac{\pi|\omega|}{2\gamma}, \quad M_\gamma(0) = 1.$$

Small gamma suppresses rapid variation. First average over continuous centers. Let Q have orthonormal columns spanning zero-mean sample patterns: $Q^T Q = I$, $Q^T \mathbf{1} = 0$. Removing the constant direction makes the whole-line center integral finite. With $t_\gamma(c) = (\tanh(\gamma(x_a - c)))_a$, $d_{ab} = x_a - x_b$,

$$H_\gamma = \frac{1}{hm} \int_{\mathbb{R}} Q^T t_\gamma(c) t_\gamma(c)^T Q dc = -\frac{2}{hm} Q^T [d_{ab} \coth(\gamma d_{ab})]_{a,b} Q.$$

For any v , its response is

$$v^T H_\gamma v = \frac{2}{\pi hm} \int_{\mathbb{R}} \frac{M_\gamma(\omega)^2}{\omega^2} \left| \sum_a (Qv)_a e^{-i\omega x_a} \right|^2 d\omega.$$

Every term is nonnegative, and increasing gamma increases the multiplier. Thus $H_\Gamma \succeq H_\gamma$ for $\Gamma \geq \gamma$: every ordered integral matrix eigenvalue increases, while its eigenvectors may change. Gamma's influence is explicit before any spectral calculation.

The actual centers form a finite lattice. Subtract the outer products of a retained set $\mathcal{F}$ of absent exterior centers:

$$T_\gamma = \frac{1}{m} \sum_{c \in \mathcal{F}} Q^T t_\gamma(c) t_\gamma(c)^T Q, \quad S_\gamma = H_\gamma - T_\gamma.$$

Appendix A bounds the entire lattice-versus-integral discrepancy by δ_γ and the remaining exterior-center tail by τ_γ :

$$S_\gamma - (\delta_\gamma + \tau_\gamma)I \preceq Q^T K_\gamma Q \preceq S_\gamma + \delta_\gamma I.$$

These allowances follow from analytic tails, rather than measured differences from the actual small eigenvalues.

Lemma (finite eigenvalue-ratio interval). Let $\beta_1 \geq \dots \geq \beta_{m-1}$ be the eigenvalues of S_γ , and let $0 < \ell_\gamma \leq \mu_1(K_\gamma) \leq L_\gamma$. For $2 \leq i \leq m-1$,

$$\underbrace{\frac{[\beta_i - \delta_\gamma - \tau_\gamma]_+}{L_\gamma}}_{\rho_i} \leq \frac{\mu_i(K_\gamma)}{\mu_1(K_\gamma)} \leq \underbrace{\min\left\{1, \frac{\beta_{i-1} + \delta_\gamma}{\ell_\gamma}\right\}}_{\bar{\rho}_i}.$$

Here $[a]_+ = \max(a, 0)$. Both endpoints are one for $i = 1$; for $i = m$, use lower endpoint zero and upper endpoint $\min\{1, (\beta_{m-1} + \delta_\gamma)/\ell_\gamma\}$. Known exact zero eigenvalues can have both endpoints zero.

The one-index shift accounts for the constant direction removed by Q ; actual eigenvectors need not have zero mean. All quantities use the current gamma and geometry. Computing the spectrum of S_γ is part of the calculation. Finite corrections and normalization can prevent individual finite ratios from increasing, even though the integral contribution increases.

3 The theorem predicts remaining output error

Theorem (frozen-feature GD output-error lower bound). Under the setup above, let $y \neq 0$, initialize $w_0 = 0$, and use $\eta = 1/(2\mu_1(K_\gamma))$. With the lemma's upper rate endpoints $\bar{\rho}_i(\gamma)$ and actual target weights $p_i(\gamma) = |u_i(\gamma)^T y|^2 / \|y\|^2$, every integer $n \geq 0$ satisfies

$$\frac{\|J_\gamma w_n - y\|^2}{\|y\|^2} \geq \underbrace{\sum_i p_i(\gamma) \left(1 - \frac{\bar{\rho}_i(\gamma)}{2}\right)^{2n}}_{e_{\text{lower}}(n)^2}.$$

The prediction uses $\bar{\rho}_i$, constructed from S_γ and its allowances. Using actual ρ_i instead gives the separate exact-spectrum reference. Target weights come from the actual finite-feature SVD; substituting integral eigenvectors would require another argument. Neither calculation uses a GD trajectory.

Proof. Expand the residual in actual kernel eigenvectors. Since $\rho_i \leq \bar{\rho}_i \leq 1$, each exact decay factor is at least its bounded counterpart. Multiply by $p_i \geq 0$ and sum: target energy in directions with small bounded rates must persist. $\square$

Necessary time. Let $n_{\text{necessary}} = \min\{n : e_{\text{lower}}(n) \leq \varepsilon\}$. The actual crossing cannot precede this count. Appendix A derives the companion upper error curve and sufficient time from lower rate endpoints.

Same-geometry validation. We use $W = 559$, $h = 1/256$, $m = 8,193$ uniform samples on $[-1, 1]$, and $f(x) = \sin(2\pi x) + \frac{1}{2} \sin(6\pi x) + \frac{1}{4} \sin(10\pi x)$. At gammas 8, 12, 16, and 64, executed GD reaches 1% after **15,798,313; 186,057; 61,792; and 16,013 updates**. The actual-spectrum formula reproduces these crossings. Figure 1 compares with the recomputed theorem intervals; Appendix B shows the curves producing the endpoints. Archived steps replace $\rho_i/2$ by $\eta\mu_1\rho_i$.

Quantitative timing check. The theorem's lower error curves first reach 1% at **13,934,809; 163,954; 54,453; and 14,118 updates**, respectively, retaining **88.1–88.2%** of the observed delay. Companion sufficient bounds range from **2.07 times** the observed count at gamma 8 to **1.27 times** at gamma 64. These are checked FP64 evaluations, with numerical allowances described in Appendix B.

Adam is an empirical comparison. At gamma 8, modes with $0 < \rho_i \leq 2 \times 10^{-6}$ initially contain **4.33%** of target energy and retain 1.47094×10^{-4} after 200,000 Adam updates, exceeding the 10^{-4} squared-error budget for 1% accuracy. Larger gammas' band energies are below budget. Adam does not obey the GD decay law. A shared 100-update half-life gives first loss-EMA crossings at **7,688; 1,900; and 1,486** for gammas 12, 16, and 64; gamma 8 never crosses within 200,000. Crossings can recur. Initial-loss memory sets a **1,329-update** floor, compressing differences between fast runs.

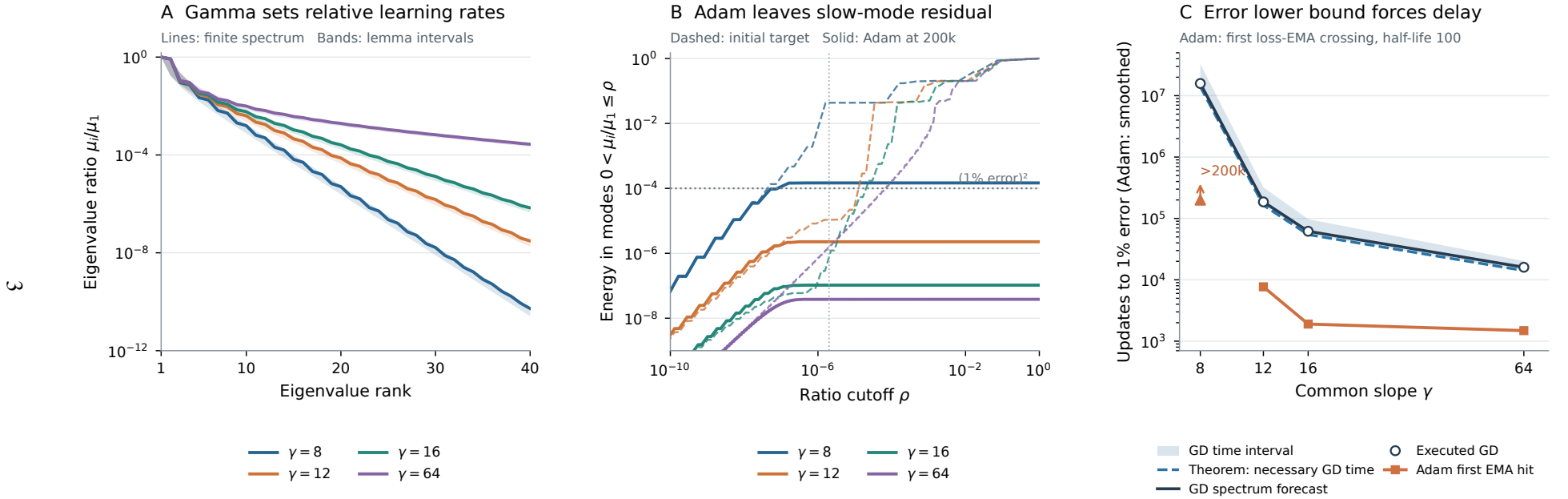


Figure 1: Gamma, needed directions, and acquisition. A: actual eigenvalue ratios and lemma enclosures at the four slopes. B: cumulative initial target energy (dashed) and Adam residual energy at 200,000 updates (solid); the blue residual exceeds the 1% budget in slow modes alone. C: the dashed theorem curve marks first crossings of the output-error lower bound; shading extends to the companion sufficient time. The solid forecast uses actual eigenvalues; circles mark executed GD. Adam markers show first EMA acquisition, with censoring at 200,000; GD has a longer budget. All panels use the same target and geometry. Bounds are checked FP64 evaluations.

Appendix A. Construction and full proof

Integral and smoothing. Write $d = x_a - x_b$. The identity $1 - \tanh u \tanh v = \coth(u - v)(\tanh u - \tanh v)$ gives

$$\int_{\mathbb{R}} [1 - \tanh(\gamma(x_a - c)) \tanh(\gamma(x_b - c))] dc = 2d \coth(\gamma d),$$

with limit $2/\gamma$ at $d = 0$. For $u \perp \mathbf{1}$, summing against $u_a u_b$ cancels the constant term and proves the formula for H_γ .

The functions $s_\gamma * \text{sign}$ and $\tanh(\gamma \cdot)$ both vanish at zero and have derivative $2s_\gamma$. With convention $\widehat{f}(\omega) = \int f(x) e^{-i\omega x} dx$, substitute $z = e^{2\gamma x}$, $a = \omega/(2\gamma)$:

$$\widehat{s}_\gamma(\omega) = \int_0^\infty \frac{z^{-ia}}{(1+z)^2} dz = \Gamma(1-ia)\Gamma(1+ia) = \frac{\pi a}{\sinh(\pi a)} = M_\gamma(\omega).$$

The beta integral and gamma-function reflection identity give the middle equalities. For $u = Qv$, $g_{\text{step}}(c) = \sum_a u_a \text{sign}(c - x_a)$ is compactly supported and has derivative $2 \sum_a u_a \delta_{x_a}$. Hence $\widehat{g}_{\text{step}} = 2 \sum_a u_a e^{-i\omega x_a} / (i\omega)$ and $\widehat{g}_\gamma = M_\gamma \widehat{g}_{\text{step}}$. Parseval proves the Fourier identity, including its prefactor. The numerator vanishes at zero because $\sum_a u_a = 0$. Since $d \log(z/\sinh z)/dz = 1/z - \coth z < 0$, increasing gamma increases M_γ . Quadratic-form order and min-max prove ordered growth of the integral eigenvalues.

Lattice allowance. Let the full center lattice be $c_0 + h\mathbb{Z}$. We retain no explicit Fourier aliases: the $p = 0$ case of V4. Choose $0 < \vartheta < \pi/2$, write $\bar{x} = m^{-1} \sum_a x_a$, and put

$$\delta_\gamma = \frac{8\gamma \|x - \bar{x}\mathbf{1}\|^2 \sec^4 \vartheta}{3hm[\exp(2\pi\vartheta/(\gamma h)) - 1]}.$$

For $u \perp \mathbf{1}$, set $g(z) = \sum_a u_a \tanh(\gamma(z - x_a))$. Subtract the common translate at $\bar{x}$, express each difference as an integral of its derivative, and apply Minkowski and Cauchy-Schwarz. On $\text{Im } z = \pm\vartheta/\gamma$ this yields

$$\int_{\mathbb{R}} |g(t \pm i\vartheta/\gamma)|^2 dt \leq \frac{4\gamma}{3} \sec^4 \vartheta \|x - \bar{x}\mathbf{1}\|^2 \|u\|^2.$$

Here $|\text{sech}^2(t + i\vartheta)| \leq \sec^2 \vartheta \text{sech}^2 t$ and $\int \text{sech}^4 t dt = 4/3$. Shift the contour for the analytic function $g(z)^2$, bounding its absolute integral by the displayed inequality. Its Fourier transform decays as $\exp(-\vartheta|\omega|/\gamma)$. Poisson summation at $2\pi k/h$ and the geometric sum over both signs give the norm bound δ_γ for the infinite-lattice minus integral matrix.

Exterior allowance. Let $\mathcal{F}$ contain a finite set of missing lattice centers, including any inside the sample range. Suppose remaining right and left tails start at $c_R > x_{\max}$ and $c_L < x_{\min}$. Then

$$\tau_\gamma = \frac{4[e^{-4\gamma(c_R - x_{\max})} + e^{-4\gamma(x_{\min} - c_L)}]}{1 - e^{-4\gamma h}}.$$

Because $Q^T \mathbf{1} = 0$ and $1 - \tanh t \leq 2e^{-2t}$ for $t \geq 0$, a right-tail projected outer product divided by m has norm at most $4e^{-4\gamma(c - x_{\max})}$. The left side is analogous.

Summing the geometric tails gives $0 \preceq T_{\text{tail}} \preceq \tau_\gamma I$. The finite kernel equals the infinite lattice minus all absent centers:

$$Q^T K_\gamma Q = S_\gamma + R_{\text{lattice}} - T_{\text{tail}}, \quad \|R_{\text{lattice}}\| \leq \delta_\gamma.$$

This proves the two-sided matrix enclosure. The bias vanishes under Q .

Eigenvalues and normalization. If κ_i are the descending eigenvalues of $Q^T K_\gamma Q$, then $\beta_i - \delta_\gamma - \tau_\gamma \leq \kappa_i \leq \beta_i + \delta_\gamma$. Interlacing gives $\mu_i \geq \kappa_i \geq \mu_{i+1}$, proving the numerator endpoints. Divide lower numerators by L_γ and upper numerators by ℓ_γ to obtain the lemma.

For the normalization used in the calculation, let $e = \mathbf{1}/\sqrt{m}$:

$$\ell_\gamma = \|J_\gamma^T e\|^2, \quad b = \|(I - ee^T)J_\gamma(J_\gamma^T e)\|, \quad c = \beta_1 + \delta_\gamma,$$

$$L_\gamma = \frac{\ell_\gamma + c + \sqrt{(\ell_\gamma - c)^2 + 4b^2}}{2}.$$

The first is a Rayleigh lower bound. In the basis $[e, Q]$, the remaining block is bounded by cI and the coupling has norm b , so the largest eigenvalue is at most that of the displayed two-by-two scalar comparison. Numerical upper allowances are also included in c . These quantities use feature matrix-vector products and the corrected integral spectrum, without the actual small eigenvalues.

GD error and times. Expanding the zero-initialized residual in the actual eigenvectors gives the error formula. Monotonicity in each ratio proves the output-error lower bound. Lower rate endpoints analogously give the companion upper bound:

$$e_{\text{lower}}(n)^2 \leq e(n)^2 \leq \underbrace{\sum_i p_i (1 - \rho_i/2)^{2n}}_{e_{\text{upper}}(n)^2}, \quad n_{\text{necessary}} \leq n_\varepsilon \leq n_{\text{sufficient}}.$$

The curves decrease with update count, so their first crossings bracket the true crossing; an empty crossing set means $+\infty$. For a general nonoscillatory step, replace $\rho/2$ by $\eta\mu_1\rho$ with $0 < \eta\mu_1 \leq 1$. A positive mode is representable because $u_i = J_\gamma(J_\gamma^T u_i/\mu_i)$. Thus target energy in slow positive modes describes optimization delay separately from nullspace capacity. $\square$

Appendix B. Numerical bounds and optimizer checks

Evaluation. The positive center integral is evaluated by Gauss–Legendre feature quadrature, followed by subtraction of retained exterior-center outer products. A shared low-rank feature basis avoids forming an $m \times m$ kernel. If $Z = [A, C]$ contains integral and exterior factors, truncating after singular value σ_r changes $Z \text{diag}(I, -I)Z^T$ by at most $2\sigma_1\sigma_{r+1} + \sigma_{r+1}^2$ in norm; this allowance is included on both sides. Pad the restricted spectrum to dimension $m - 1$, placing zeros between positive and negative reduced eigenvalues.

Truncating the whole-line integral at $\pm(\max_a |x_a| + P/\gamma)$ omits a positive matrix of norm at most $2e^{-4P}/(h\gamma)$, included on the upper side. The remaining exterior tail contributes on the lower side. Quadrature refinement and FP64 sensitivity are checked separately from analytic allowances. Reported numerical intervals are not directed-rounding certificates.

Use order-10 quadrature on center panels of width at most $1/\gamma$, $P = 20$, $\vartheta = \arctan(\pi/(2\gamma h))$, and $\lceil 18/(\gamma h) \rceil$ retained exterior centers per side. The feature-factor cutoff is 10^{-14} relative to its largest singular value. An empirical arithmetic allowance $64\epsilon_{64}\|S_{\text{reduced}}\|$ is added on both sides. Refining to order 16 and $P = 24$ preserves every integer time endpoint; spectrum changes are below the combined compression and arithmetic allowances. All resolved ratios and sampled error curves remain enclosed. The analytic lattice allowance ranges from 4.37×10^{-126} at gamma 8 to 9.02×10^{-9} at gamma 64, in absolute kernel units.

Actual finite-feature SVD supplies the target weights and comparison spectrum. Ratios at most 10^{-18} are treated as numerically unresolved: their energy is retained in the upper error curve and omitted only from the lower curve. It is tracked separately from the directly computed projection residual. Powers use $\log 1p$; integer crossings use bracketing and binary search. No GD or Adam training is rerun.

Theorem rate caps predict a lower bound on output error

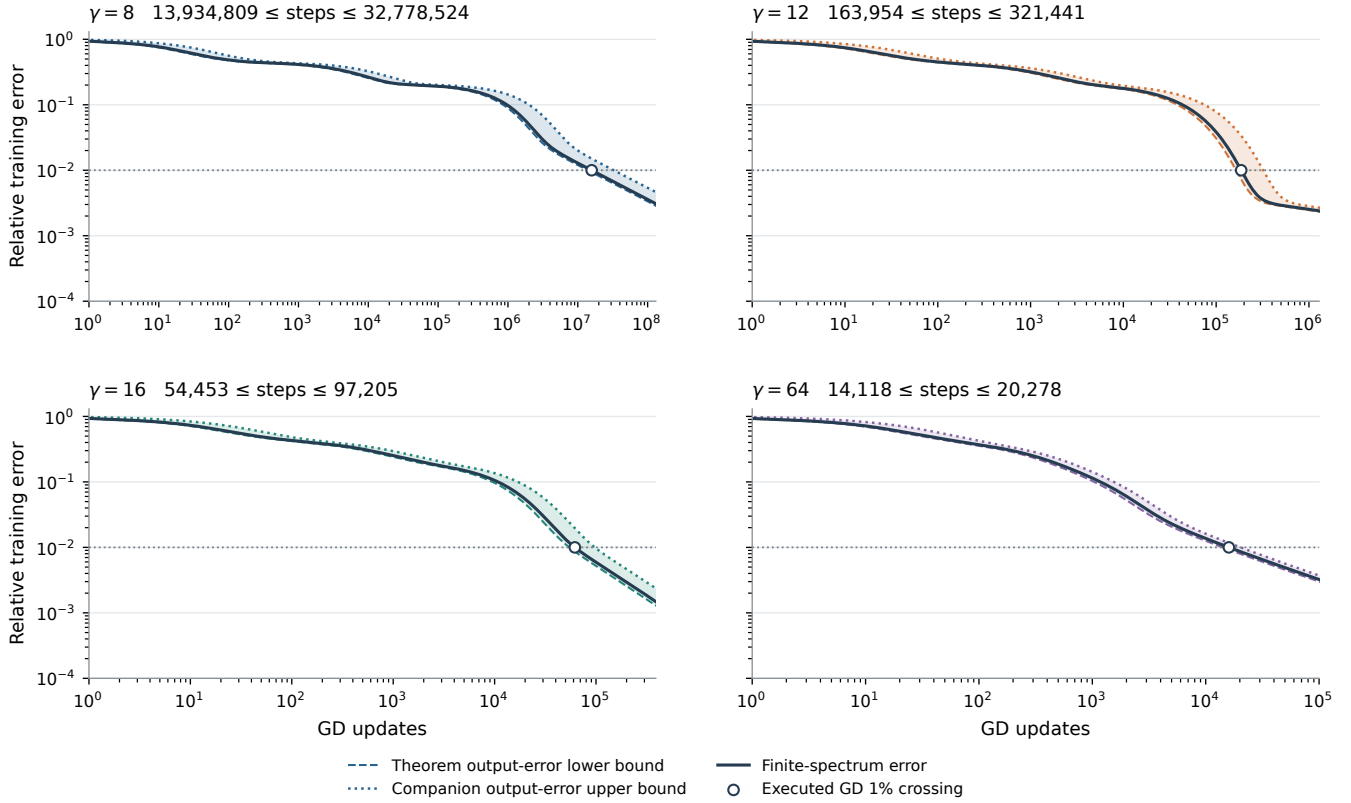


Figure 2: The theorem predicts remaining output error. The dashed lower bound substitutes the lemma's upper rate endpoints into the GD formula; its 1% crossing gives the necessary time. The dotted companion upper bound uses lower rate endpoints. The solid reference uses actual eigenvalues; the marked crossing comes from executed GD. All curves retain the same actual target weights, with unresolved energy handled conservatively.

Adam protocol. The common setting uses initial rate 10^{-3} , epsilon 10^{-12} , and moments (0.9, 0.999). The rate is fixed through 20,000 updates, decays by cosine to 10^{-6} at 50,000, then remains fixed through 200,000. Define

$$M_0 = 1, \quad M_n = \beta M_{n-1} + (1 - \beta)e(n)^2, \quad \beta = 2^{-1/100}.$$

First EMA acquisition is the first $M_n \leq 10^{-4}$, computed from every update. Since $M_n \geq \beta^n$, no crossing occurs before 1,329 updates. Later upcrossings occur 73, 97, and 98 times at gammas 12, 16, and 64; sustained raw crossings occur around 41,000.



Figure 3: Adam’s measured acquisition. Bands show raw-error minima and maxima within update bins; curves show the square root of the loss EMA. Dashed verticals mark first crossings; gray windows mark rate decay. Display sampling does not affect the every-update crossing calculation.

The common-setting ordering persists at EMA half-lives 30, 100, and 300, but changes at longer windows and for validation-selected settings. The archived [five-target comparison](#) and [window sensitivity](#) show this dependence. Adam projections measure current error and can increase during training; they are diagnostics, not a GD rate law.

Reproduction. The [ratio calculation](#) and its [evidence directory](#) record geometry, analytic and numerical allowances, refinement checks, source hashes, and crossings. [Optimizer evidence](#) retains modal projections and EMA measurements. All results concern training-grid optimization with frozen features.